



The Omics Hub

BIOINFORMATICS MASTER CHEAT SHEET

1. Command Line & Bash Core

File & Directory Operations

<code>pwd</code>	Print current working directory
<code>ls -lah</code>	List all files with human-readable sizes
<code>cd /dir/</code>	Change directory
<code>mkdir -p</code>	Create directory and parents if needed
<code>cp -r</code>	Copy files or directories recursively
<code>mv</code>	Move or rename files
<code>rm -rf</code>	Remove files recursively (Use with caution!)

File Inspection & Text Parsing

<code>head -n 20</code>	View the first 20 lines of a file
<code>tail -f</code>	Output appended data as the file grows
<code>less -S</code>	View file, do not wrap long lines
<code>wc -l</code>	Count the total number of lines in a file
<code>grep -i</code>	Search for a pattern (case-insensitive)
<code>grep -v</code>	Invert match (exclude lines matching pattern)
<code>awk '{print \$1}'</code>	Print only the 1st column of a space-delimited file

Data Compression & Transfer

<code>tar -czvf</code>	Compress directory into a .tar.gz archive
<code>tar -xzf</code>	Extract a .tar.gz archive
<code>gzip file</code>	Compress a file (creates file.gz)
<code>zcat file.gz</code>	View contents of a gzipped file without extracting
<code>rsync -avz</code>	Securely copy/sync files between servers
<code>scp file user@host:</code>	Securely copy file to a remote server

System & Job Management (Slurm)

<code>htop</code>	Interactive process viewer
<code>df -h</code>	View disk space usage
<code>du -sh *</code>	Check folder sizes in current directory
<code>squeue -u \$USER</code>	View your active Slurm jobs
<code>scancel <jobid></code>	Cancel a specific Slurm job
<code>sbatch script.sh</code>	Submit a job to the cluster

```
sed 's/old/new/g'
```

Replace "old" with "new"
globally in text

Bash Scripting Best Practices

Always start scripts with `#!/bin/bash` . Use `set -euo pipefail` to make your script exit on errors, undefined variables, or pipeline failures. Quote your variables (e.g., `"$FILE"`) to prevent word splitting.

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2. Conda / Mamba Environment Management

Conda (and its faster drop-in replacement, Mamba) is essential for maintaining reproducible software environments in bioinformatics.

Environment Creation & Activation

<code>conda create -n env_name</code>	Create a new, empty environment
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<code>conda create -n env python=3.9</code>	Create environment with specific Python version
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<code>conda activate env_name</code>	Activate the environment
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<code>conda deactivate</code>	Deactivate current environment
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<code>conda env list</code>	List all installed environments
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<code>conda remove -n env --all</code>	Completely delete an environment
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Package Management

<code>conda install pkg</code>	Install a package in active env
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<code>conda install -c bioconda</code>	Install package from Bioconda channel
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<code>conda search pkg</code>	Search for available versions of a package
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<code>conda list</code>	List installed packages in active env
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<code>conda update pkg</code>	Update a specific package
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<code>conda clean --all</code>	Remove unused packages and caches (free up disk)
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Environment Export & Reproducibility

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# Export the current environment to a YAML file for sharing
conda env export > environment.yml
```

```
# Create a new environment from a shared YAML file
conda env create -f environment.yml
```

3. R & Seurat Core Functions

Tidyverse Data Manipulation

<code>filter(df, condition)</code>	Keep rows that match a condition
<code>select(df, col1, col2)</code>	Keep only specific columns
<code>mutate(df, new=x+y)</code>	Create or modify columns
<code>group_by(df, col)</code>	Group data by a categorical variable
<code>summarize(df, m=mean(x))</code>	Calculate summary statistics per group
<code>left_join(x, y, by="id")</code>	Merge two dataframes by a common key

ggplot2 Visualization

<code>ggplot(df, aes(x, y))</code>	Initialize a ggplot object
<code>geom_point()</code>	Add a scatterplot layer
<code>geom_boxplot()</code>	Add a boxplot layer
<code>facet_wrap(~ group)</code>	Create multi-panel plots by group
<code>theme_minimal()</code>	Apply a clean, minimal theme
<code>ggsave("plot.pdf")</code>	Save the last generated plot

Standard Seurat Workflow (scRNA-seq)

```
# 1. Initialize and Filter
seurat_obj <- CreateSeuratObject(counts = raw_data, project = "Sample1", min.cells = 3, min.features = 200)
seurat_obj[["percent.mt"]] <- PercentageFeatureSet(seurat_obj, pattern = "^MT-")
seurat_obj <- subset(seurat_obj, subset = nFeature_RNA > 200 & nFeature_RNA < 2500 & percent.mt < 5)

# 2. Normalize and Identify Highly Variable Features
seurat_obj <- NormalizeData(seurat_obj)
seurat_obj <- FindVariableFeatures(seurat_obj, selection.method = "vst", nfeatures = 2000)

# 3. Scale Data and Run PCA
seurat_obj <- ScaleData(seurat_obj)
seurat_obj <- RunPCA(seurat_obj, features = VariableFeatures(object = seurat_obj))

# 4. Clustering and UMAP
seurat_obj <- FindNeighbors(seurat_obj, dims = 1:15)
seurat_obj <- FindClusters(seurat_obj, resolution = 0.5)
seurat_obj <- RunUMAP(seurat_obj, dims = 1:15)

# 5. Visualization & Marker Identification
DimPlot(seurat_obj, reduction = "umap", label = TRUE)
FeaturePlot(seurat_obj, features = c("CD3D", "CD19", "MS4A1"))
markers <- FindAllMarkers(seurat_obj, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

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4. Genes for scRNA-seq Annotation

Standardized marker genes used for cell-type identification in single-cell transcriptomics, with a specialized focus on T-cell immunology, the tumor microenvironment, and Sézary Syndrome.

Broad Immune Compartments (PBMC / TME)

Cell Type	Canonical Marker Genes
T Cells (Pan)	CD3D , CD3E , CD3G , TRAC , IL32
B Cells	CD19 , MS4A1 (CD20) , CD79A , CD79B
Plasma Cells	SDC1 (CD138) , MZB1 , JCHAIN , IGKC
NK Cells	NCAM1 (CD56) , KLRD1 (CD94) , NKG7 , GNLY , FCGR3A (CD16)
Monocytes (Classical)	CD14 , LYZ , VCAN , S100A9 , S100A8
Monocytes (Non-classical)	FCGR3A (CD16) , MS4A7 , LST1
Macrophages	CD68 , CD163 , MRC1 (CD206) , C1QA
Dendritic Cells (cDC)	HLA-DRA , CST3 , CLEC9A (cDC1), CD1C (cDC2)
pDC (Plasmacytoid)	LILRA4 , CLEC4C (CD303) , TCF4

Detailed T-Cell Subsets & Exhaustion

Sub-phenotype	Canonical Marker Genes
CD4+ Helper T Cells	CD4 , IL7R , CCR7 (Naïve/Tcm), SELL (CD62L)
CD8+ Cytotoxic T Cells	CD8A , CD8B , GZMB , PRF1 , IFNG
Regulatory T Cells (Treg)	FOXP3 , IL2RA (CD25) , CTLA4 , TIGIT , IKZF2 (Helios)
Th1 Cells	TBX21 (T-bet) , IFNG , CXCR3
Th2 Cells	GATA3 , IL4 , IL5 , IL13 , PTGDR2 (CRTH2)
Th17 Cells	RORC , IL17A , CCR6 , KLRB1 (CD161)
Exhausted T Cells (Tex)	PDCD1 (PD-1) , HAVCR2 (TIM-3) , LAG3 , CTLA4 , TOX

Sézary Syndrome & Advanced Biomarkers

Signatures specifically enriched or altered in CTCL / Sézary Syndrome profiling.

Pathway / Relevance	Canonical Marker Genes
Malignant Clone Hallmarks	TOX (overexpressed), PLS3 (Plastin 3), KIR3DL2 (CD158k)
Loss of Surface Antigens	Downregulation/Loss of CD7 and CD26 (DPP4) is typical of SS cells
Skin Homing Receptors	CCR4 (target for Mogamulizumab), CLA
Metabolic Dysregulation	CLIC1 (Chloride Intracellular Channel 1 - linked to oxidative stress)
Malignant vs Benign Treg	Distinguishing suppressive subsets requires evaluating FOXP3 and CD25 expression profiles compared to the malignant clonal population.

Important Note on Annotations

Gene expression can vary highly based on the tissue state (steady-state vs. inflamed vs. tumor). Always validate computational annotations by examining the expression of a combination of markers via FeaturePlot or DotPlot rather than relying on a single gene.